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Group 7
Earthquake & Tsunami Prediction



1. Abstract

This report explores the **prediction of tsunami occurrence** and **earthquake magnitudes** through a machine learning framework grounded in geophysical and oceanographic features. Our study integrates robust data preprocessing and feature engineering to enable forecasts. For tsunami classification, we ultimately opted for **XGBoost** with tuned hyperparameters, focusing on **maximising recall** to ensure all tsunami events are captured, while minimising **false positives** to reduce unnecessary alarms. For earthquake magnitude regression, **XGBoost** with the log-transformed target variable proved most effective in **minimising error**, especially for high-magnitude events. Together, these models offer a complementary risk assessment tool to support government disaster response planning and insurance underwriting decisions for mitigating damage and losses.

We used three datasets: (1) global earthquake-tsunami data from 1995-2023 from Kaggle (Kaggle, 2023), (2) NOAA's tsunami data (NOAA, 2023), and (3) a wikipedia summary table to build a unified view of seismic activity. To address documentation inconsistencies, particularly between the pre- and post-2004 DART deployment, we performed extensive cleaning and feature engineering. **Class imbalance** was addressed using **SMOTE**, and models were evaluated using **cross-validation** and appropriate metrics for each predictive task.

In conclusion, the study developed effective predictive models for both tsunami classification and earthquake magnitude regression. The classification model achieved high recall, successfully identifying the vast majority of tsunami events caused by earthquakes, while balancing the trade-off with false positives. Simultaneously, the regression model attained a mean absolute error of 0.1, which is crucial for anticipating seismic events of varying severities.

2. Introduction

2.1 Background and Context

Earthquakes are powerful and unpredictable natural hazards that can cause widespread destruction within seconds. Beyond ground shaking, they may also trigger secondary disasters such as tsunamis, especially when undersea seismic activity displaces large volumes of water. Tsunamis, though often undetectable in deep water, gain destructive force near coastlines, leading to catastrophic flooding and loss of life. While not all earthquakes generate tsunamis, those that do often share specific traits. Identifying these patterns is essential for improving early warning systems and reducing risk to coastal populations. Understanding the link between seismic activity and tsunami formation has long been a focus in geophysics and disaster management. Insights from past events can enhance forecasting, inform urban planning, and strengthen emergency response in vulnerable regions.

2.2 Problem Statement

This study leverages predictive analytics to explore the relationship between seismic characteristics, geophysical data, and disaster outcomes, with a focus on enhancing early warning systems and emergency preparedness. Recognising that approximately 89% of tsunamis are triggered by earthquakes (Tsunami Generation: Earthquakes, 2023), it becomes crucial to understand the factors that influence both earthquake severity and tsunami generation.

Our objective is to develop robust machine learning models that can (1) accurately estimate earthquake magnitudes using a regression model and (2) predict tsunami occurrence using a classification model. By analyzing historical seismic data, we aim to uncover patterns and key indicators that can help differentiate between low and high-risk events. This research seeks to equip disaster response agencies with more accurate and timely insights, ultimately supporting more effective risk assessment, prioritisation of emergency resources, and protection of vulnerable communities

2.3 Data

This study utilizes an integrated dataset from three primary sources to support tsunami classification and earthquake magnitude regression tasks. The core dataset, sourced from Kaggle, covers global earthquake and tsunami events from 1995 to 2023. Due to limited documentation and unknown sourcing, additional records from the NOAA Earthquake and Tsunami Databases, which are maintained by the National Centers for Environmental Information (NCEI), were incorporated to enhance reliability. These official datasets improve contextual depth but are noisier before 2004, prior to the widespread deployment of the DART (Deep-ocean Assessment and Reporting of Tsunamis) system. Consequently, pre-2004 entries were filtered out. To further validate coverage of major events, a summary table from Wikipedia was used for cross-referencing.

Key variables were grouped into four categories: temporal, seismic, geographic, and impact-related. From the Kaggle dataset, we retained: “date_time”, “magnitude”, “cdi”, “mmi”, “nst”, “tsunami”, “continent”, and “country”. From the NOAA Tsunami Database we mainly used: “Year”, “Mo”, “Dy”, “Hr”, “Mn”, “Sec”, “Earthquake Magnitude”, “Country”, and “Location Name”. Full definitions are provided in Appendix 1.

2.4 Methods

Our predictive modeling process adheres to a **structured pipeline**: data preprocessing, conditional resampling, feature engineering, model training, and hyperparameter optimization, all embedded within a **stratified K-fold cross-validation** framework to ensure statistical robustness. Given the limited sample size, stratification preserved the target distribution across folds, mitigating variance and potential overfitting. We evaluated a suite of models including **Random Forest**, **XGBoost**, **K-Nearest Neighbors (KNN)**, and **Support Vector Regression (SVR)**, chosen for their effectiveness in small-sample, tabular data contexts. To address the skewness in the regression target variable, we implemented XGBoost with quantile regression, enabling the estimation of conditional quantiles rather than point predictions. Model selection and tuning were conducted via **grid and randomized search**, optimizing performance based on metrics such as **MSE**, **MAE**, and **R²**.

3. Data

To create a robust dataset for tsunami classification and earthquake magnitude regression, we have integrated four primary data sources. This unified dataset merges technical, geographic, and impact-based data, addressing existing documentation issues and offering a comprehensive view of tsunami and earthquake events.

1. [Kaggle Earthquake-Tsunami Dataset \(1995–2023\)](#) (Kaggle, 2023):
This tabular data in the csv format (1,000 rows with 18 features) includes detailed seismic attributes such as magnitude, depth, tsunami cause codes, geographic coordinates, and wave characteristics. It served as the core dataset due to its granularity in geophysical variables relevant to tsunami formation.
2. [NOAA Tsunami Database](#) (NOAA, 2022):
This tabular data contains tsunami events officially documented by NOAA, with selected seismic features.
3. [Wikipedia Summary Table \(List of Tsunamis\)](#) (Wikipedia, 2025):
This HTML table provides a curated summary of major tsunami events, particularly from 1950 to 2000. While less granular, it helps cross-reference key events and validate coverage, especially for landmark disasters.

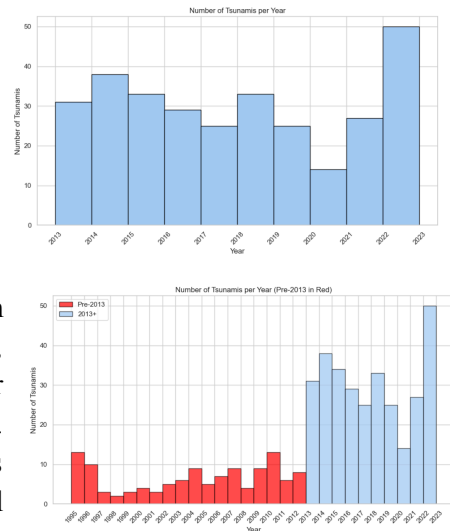
3.1 Imputing Missing, Mislabeled and Wrong Data

3.1.1 Imputing Missing Geographical Data

To impute missing 'country' and 'continent' values, we extracted country names from the 'location' column and applied custom string cleaning rules to standardise naming. For continent assignment, we used 'pycountry_convert' to map standardised country names to continent codes. Records that could not be mapped due to ambiguous or overly broad islandic regions were dropped to maintain geographical precision.

3.1.2 Imputing Missing Tsunami Data

From 1995 to 2013, many events were mistakenly recorded with a 0 label (no tsunami) instead of null, masking missingness. To correct mislabeled tsunami occurrences from 1995 to 2013, we leveraged the NOAA tsunami database and the Wikipedia HTML table of historical tsunami events. We applied a **fuzzy-matching** approach based on earthquake magnitude, cause, timestamp, and location keywords to identify matching tsunami events in both datasets. While this recovered key tsunami events, the number of positive cases from this period remained low. Given that DART tsunami detection systems were only deployed post-2004 (Sim, 2014), and data before that remains sparse even after enrichment, we restricted our classification task to events from 2004 onwards, where labels are more trustworthy. We also introduced a pre_2013_era feature to explicitly flag lower-quality records from 1995–2012 during modeling, allowing the model to account for residual label bias.



3.1.3 Imputing Zero Values in Numeric Features

The 'nst', 'dmin', 'cdi', and 'gap' columns contained numerous instances of zero values. While individually these might occur under specific conditions, rows where all four of these features were simultaneously zero are highly suspect, as a seismic event cannot be recorded without any station detection (nst), distance to epicenter (dmin), perceived intensity (cdi), or azimuthal gap coverage (gap). These cases likely reflect missing or corrupted data and were removed entirely to preserve the quality and physical plausibility of the dataset. However, zero values in individual columns still remained. Since feature importance analysis showed that 'nst', 'dmin', and 'gap' retained slight predictive power for the target variables (information value > 0.01), we opted to impute them instead of discarding further rows. We applied a hierarchical **median** imputation strategy, first using fine-grained geographic groupings (latitude, longitude, and depth), then broader ones (location-depth and country-depth), and finally resorting to global medians for unresolved cases. For 'cdi', which represents community-reported intensity, we observed that its values typically exceed magnitude. We modeled the distribution of the difference (cdi - magnitude) using a log-normal distribution fitted to valid cases, then sampled from this to impute missing CDI values by adding sampled values to the corresponding magnitude and bounding results within the realistic range of 1 to 9. This approach aims to maintain data coherence while preserving meaningful variability and relationships in the input features.

3.2 Handling High-Dimensionality of Data

3.2.1 Seismic Groupings for 'country' and Binary 'Island' Column

The 'country' feature contains 57 unique values, which could contribute to high dimensionality after encoding. It was mapped to seismic region groupings based on known tectonic zones. Countries were assigned to specific seismic regions such as the "Ring of Fire," "Himalayan Belt," and "Mediterranean-Asian Belt," reflecting their geological settings and seismic activity. This grouping **reduces the cardinality** of the 'country' feature and helps capture relevant patterns associated with regional seismicity. However, this process risks omitting meaningful nuances that could influence tsunami risk. For instance, island nations often experience different levels of tsunami exposure compared to inland countries, due to their proximity to tectonic plate boundaries. To capture this distinction, we engineered a binary 'Island' column to indicate whether a location is situated on an island. This feature complements seismic_region by incorporating a finer-grained geographic feature that may not be fully captured through tectonic grouping alone.

3.2.2 Magnitude Scale Groupings for 'magType'

The 'magType' feature had significantly high cardinality, as it included several different types of magnitude measurements with varying frequencies. To address this, we grouped the magnitude types into broader categories to **reduce dimensionality** without losing essential information. Specifically, all moment magnitude types such as 'mww', 'mwc', 'mwb', and 'mw' share the same fundamental measurement method based on seismic moment, so they were consolidated into a single 'Moment' category. Meanwhile, other types, 'mb', 'ms', and 'ml', which represent legacy or alternative methods, were grouped under 'Other'. This grouping reduces the number of categories while still preserving distinctions between measurement types, which may have implications for earthquake characterisation and energy release.

3.2.3 Temporal Binning of 'Year'

Given the relatively small size of the dataset (approximately 970 rows) and the presence of 29 unique years, representing each year individually would introduce high cardinality and increase the risk of overfitting. Additionally, we chose not to treat 'Year' as a continuous numerical variable, as doing so would falsely suggest a linear relationship with the target. (E.g, implying that more recent years inherently correlate with a higher likelihood of tsunamis), which lacks a theoretical basis and could bias the model. One-hot encoding the 'Year' feature would result in 29 sparse columns, many of which would have fewer than 30 samples per year, further complicating the model. Hence, we grouped the years into intervals, labeled as 'Year_Bucket' to capture meaningful temporal patterns. This grouping allows us to capture important temporal trends, without overfitting to individual years. The 'Year_Bucket' categories are now represented as 5-year intervals.

3.3 Normalising Distribution of Numerical Variables

Many numerical columns in the dataset exhibit **skewness**, which can affect the performance of certain models (Appendix 2). To address this, **log transformations** were applied to specific columns which showed significant improvement in their skewness ('sig', 'dmin', 'depth', 'gap') after applying the $\log_{1p}$ transformation (*where $\log_{1p}$ means $\log(1 + x)$*) to help in reducing the impact of large outliers. However, the 'magnitude' variable among others did not show a significant improvement with this transformation or other techniques (e.g. square root, cube root, Box-Cox). We pay special attention to the magnitude variable as it will be used as our target variable later on. Yet, the performance gains were negligible. For example, the **Mean Absolute Error (MAE)** remained around **0.256–0.257**, and **R²** showed **minimal variation** (Appendix 3). The marginal performance improvements and loss of interpretability did not justify transforming the target. Moreover, retaining magnitude in its original scale preserves its physical meaning, which may communicate the severity of events. Hence, we retained the untransformed magnitude column as the target variable.

3.4 Encoding & Standardisation

Categorical variables were **one-hot encoded** with `drop_first=True` to avoid the dummy variable trap, as they lacked ordinal meaning. Numerical features were standardised using `StandardScaler` to ensure that models relying on regularization or distance-based metrics are not skewed by differences in feature scales. This preprocessing step promotes model stability, training efficiency, and comparability across features.

4. Methodology

4.1 Classification: Tsunami Occurrence

4.1.1 Model Evaluation Strategy

In tsunami prediction, minimising false negatives is critical, as missing an actual tsunami can lead to devastating consequences. To address this, we prioritised **recall** ($\frac{TP}{TP+FN}$) as our main evaluation metric, ensuring the model captures as many true tsunami events as possible, at a reasonable cost of more false alarms (i.e., False Positives). To assess discriminatory power between the tsunami and non-tsunami classes, we also considered **AUC-ROC**, which measures how well the model separates the classes across thresholds. It is especially useful in contexts of class imbalance. Our dataset showed a moderate overall imbalance (Appendix 4). However, the pre-2013 subset, where label quality is less reliable, exhibited more imbalance: 24% of observations were labeled as tsunamis, compared to 69% in the post-2013 data. Given this, we used recall and AUC-ROC to ensure balanced model performance across thresholds and subgroups.

We used a 70-30 split for our training and test sets. This split was performed randomly to ensure that the data is representative of the overall distribution. **Stratified 5-fold cross-validation** was used on the train set to ensure that each fold contained a proportional representation of both the tsunami and non-tsunami classes. This is especially important for our dataset, which is relatively small (675 observations).

4.1.2 Resampling

Optimising Downweighting of Low Quality (Pre-2013) Data

To address the data quality issue in the pre-2013 subset, where label reliability is lower, we optimised the weight assigned to these samples using **Logistic Regression** as a base model, which was chosen for its simplicity and interpretability. We tested weights between 0.05 and 1.0 to assess their impact on model performance. Following, we evaluated the optimal weight using the best combination of **cross-validated recall** and **AUC** scores. Then, we used it to downweight the pre-2013 training data.

Class Imbalance in Pre-2013 Data

In the pre-2013 data, tsunamis are likely underreported, resulting in label 1 (tsunami) being the minority class. In contrast, tsunami is the majority class in post-2013 records (See 4.1.1). To avoid amplifying noise in the more reliable data, we applied targeted resampling only to the pre-2013 subset. **SMOTE** and **SMOTE-Tomek** were chosen. SMOTE generates synthetic minority samples, while SMOTE-Tomek also removes ambiguous boundary points to improve class separation. We aim to find the method and weight that produces the best cross-validated recall and AUC scores before applying it to the training data.

4.1.3 Feature Selection

Dropping Features with Strong Correlations

Before selecting features, we addressed **multicollinearity** in geographic variables. The ‘seismic_region’ and one-hot encoded continent columns showed strong correlation (Appendix 5), which is expected given that many tectonic plates are largely bounded by continents (e.g., the Eurasian Plate aligning with Asia/Europe). However, keeping both could introduce noise and overcomplicate interpretation. Since ‘seismic_region’ is more directly tied to tectonic activity, we dropped the continent variables to improve model clarity.

Forward Selection (AIC and BIC)

Forward selection, guided by **AIC** and **BIC**, helps balance model fit with simplicity in logistic regression. AIC is calculated as $AIC = 2k - 2\ln(\hat{L})$, where k is the number of features and $\hat{L}$ is the likelihood estimate, aiming to minimise the trade-off between goodness-of-fit and model complexity. BIC is calculated as $BIC = \ln(n)k - 2\ln(\hat{L})$ where n is the number of observations, and it imposes a stronger penalty for model complexity. In forward selection, the process starts with no features and iteratively adds those that most reduce AIC or BIC, stopping when no further improvement is observed. This avoids overfitting by selecting a model that explains the data well without becoming too complex. The final feature set corresponds to the optimal k , where AIC and BIC dropped substantially, which indicates a meaningful gain in fit without overcomplicating the model, while cross-validated recall and AUC remain high. Careful judgment is needed, as BIC is known to favour simpler models more aggressively, while AIC may retain more features at the risk of slightly increased complexity.

Lasso (L1) Regularisation

We applied logistic regression with **L1 regularisation**, which penalises the absolute magnitude of coefficients, encouraging sparsity by shrinking less informative features to zero. We tuned the inverse regularisation parameter

C using cross-validation and selected the feature set corresponding to the optimal C where the combination of cross-validated AUC and recall scores are maximised.

Random Forest Classifier

We trained a **Random Forest classifier** with default settings and extracted feature importance scores, measured by how much they reduce prediction error across the trees, averaged over multiple decision trees. This method captures nonlinear relationships and interaction effects that linear models may miss, as it splits the data based on feature thresholds that maximise class separation at each node, allowing for complex decision boundaries. We ranked the features by their importance and selected the top 15 most predictive variables.

Then, we triangulated the results from the feature selection methods to identify variables that consistently appear, which signal strong and stable predictive power across both linear and nonlinear frameworks. This helps reduce model variance and increases confidence that the retained features are genuinely informative, rather than artifacts of a specific technique.

4.1.4 Model Selection & Hyperparameter Tuning

To identify the most suitable classifier for tsunami prediction, we evaluated five models using stratified 5-fold cross-validation and grid search hyperparameter tuning. Besides L1 Logistic Regression and Random Forest which we introduced earlier, we discuss the remaining 3 models of choice:

Candidate Model/ Parameters	Strengths
K-Nearest Neighbours ‘n_neighbors’, ‘metric’	KNN is non-parametric and does not make assumptions about the data distribution. It is flexible in handling various types of relationships and can work well when data has clear local patterns or clusters, which might be the case for certain geographical or seismic regions.
SVM (Linear & RBF) ‘C’, ‘kernel’, ‘gamma’	SVM could perform well for capturing non-linear relationships in high-dimensional spaces, which is crucial when dealing with complex features like seismic_region, magnitude, and depth.
XGBoost ‘n_estimators’, ‘max_depth’, ‘learning_rate’, ‘scale_pos_weight’	XGBoost is an ensemble learning method that combines multiple decision trees to iteratively correct errors, offering high accuracy and strong handling of both linear and non-linear relationships. It integrates L1 or L2 regularisation to prevent overfitting and processes the data in a parallel learning way.

We initially considered a cost-sensitive threshold tuning approach based on estimated financial impacts from past literature: modeling the expected cost as a weighted combination of false negatives (FN) and false positives (FP), using figures such as \$5,000,000 for a missed tsunami warning and \$800,000 for an unnecessary evacuation (Cox, 1977; Nof et al., 2025). However, we ultimately chose not to rely on these specific cost values, as recent and comprehensive data on the real-world costs of tsunami-related decisions remains sparse and potentially outdated. Instead, we focused on assessing the robustness of our model’s performance under different false positive to false negative (FP:FN) **cost ratios**. For each cost ratio, we computed the expected cost at the default decision threshold (0.5) and compared it to the minimum possible expected cost achievable through threshold tuning. We then plotted the relative cost inefficiency to reveal how far the default threshold deviates from the optimal cost at each ratio. This approach provides insights into the model's behavior under varying operational priorities while avoiding over-reliance on uncertain absolute cost figures.

4.2 Regression: Earthquake Magnitude

4.2.1 Model Evaluation Strategy

We will employ Mean Absolute Error (given by $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$), Mean Squared Error (given by

$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$) and adjusted R-squared (adj. R^2) to assess the performance of the models. Since **MAE** weighs

all errors equally regardless of its size, it is suitable for the right-skewed distribution of the target variable (See 3.3; Appendix 2) as it will not focus on minimising the impact of a few extreme cases. **MSE** emphasises larger errors due to the squaring term, which is helpful for evaluating the model robustness in predicting extreme magnitudes. **Adj. R^2**

provides a measure of overall fit while accounting for the number of predictors used, helping guard against overfitting. We will evaluate the model based on the one that achieves the lowest MAE and MSE, while maintaining a high adj. R^2 .

We used a 70-30 split for our training and test sets. This split was performed randomly to ensure that the data is representative of the overall distribution. Stratified 5-fold cross-validation was used on the train set, by using the *KBinsDiscretizer* to bin the target into 5 quantiles to ensure that each fold in the cross-validation process had a proportional distribution of the target variable, which can offset the disproportionate representations of the skewed target.

4.2.2 Feature Engineering

Intentional Dropping & Adding of Features

We excluded the ‘tsunami’ variable from the predictors as it is a downstream consequence of earthquake magnitude, so including it would introduce **label leakage**. To capture physical characteristics of seismic events, we engineered an interaction feature, ‘depth_magnitude’, which is derived from the product of ‘depth’ and ‘magnitude’. This captures how deep earthquakes release energy differently than shallow ones, which can affect magnitude predictions. Next, we introduced ‘shallow_quake’ as a flag to indicate quakes with depth < 70km (log-transformed), as shallow events may behave differently and tend to cause more surface impact. To help the model capture threshold effects typical of high-energy quakes, we introduced ‘major_quake’ to indicate a flag for events with magnitude > 6.0.

RFECV

Recursive Feature Elimination with Cross-Validation is a greedy method that starts with all features and iteratively eliminates the least important ones. At each step, the model performance is evaluated. This process continues until the optimal number of features is identified, maximising the model's performance by balancing simplicity and predictive power. Unlike forward selection (as described in 4.1.3), which begins with no features and adds them one by one, **RFECV** works in reverse by eliminating features based on their importance. The optimal feature set is determined when removing further features results in a decrease in model performance.

Lasso Regularisation

We tuned the hyperparameter ‘C’ via cross-validation and selected the features that achieved the lowest combination of MAE and MSE, while maintaining the highest adj. R^2 , similar to our reasoning and approach in classification (See 4.1.3).

Random Forest Regression

Using the default parameters, we ranked features by their importance scores and retained the top 15 predictors to leverage the model’s ability to capture nonlinear relationships, similar to our reasoning and approach in classification (See 4.1.3).

4.2.3 Model Selection & Hyperparameter Tuning

To identify the most suitable regressor for earthquake magnitude, we evaluated five models and the baseline model, Linear Regression, using stratified 5-fold cross-validation and grid search hyperparameter tuning. Besides Random Forest Regression and XGBoost which we introduced earlier, we discuss the remaining three models of choice:

Candidate Model/ Parameters	Strengths
K-Nearest Neighbours (‘n_neighbors’, ‘weights’)	KNN regression predicts the target value for a given instance by averaging the target value of its “k” nearest neighbours. It is useful for capturing local relationships without assuming linearity.
SVR (‘C’, ‘kernel’, ‘epsilon’)	SVR finds the hyperplane that best fits the data while ensuring that the margin is maximised. With the kernel, it is capable of handling nonlinear relationships, so SVR is effective when the relationship between the features and target is strictly nonlinear.
XGBoost (Quantile) (‘n_estimators’, ‘max_depth’, ‘learning_rate’, ‘quantile_alpha’)	This variant of XGBoost predicts different quantiles of the target distribution, rather than the mean

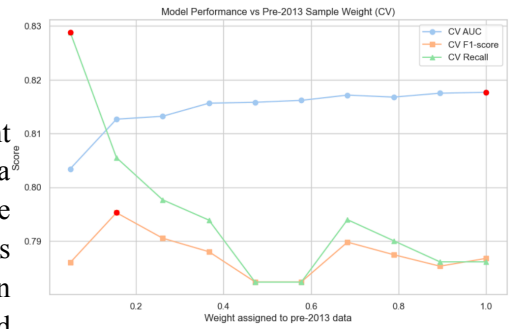
If we notice that the MAE or MSE is consistently larger for higher-magnitude earthquakes across models, it suggests that the model is not performing as well for these critical events. To address this, we assign higher weights to larger earthquakes during training, making the model more sensitive to errors in predicting them. By iteratively testing different weight values, we can find the optimal weight that minimises the errors for higher-magnitude earthquakes, thus improving the model's accuracy in predicting these important events while maintaining overall performance.

5. Results

5.1 Classification: Tsunami Occurrence

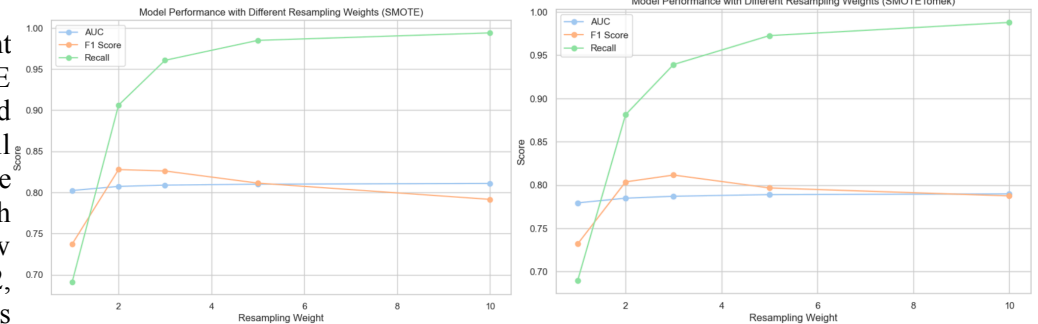
5.1.1 Selected Downweight of Low Quality (Pre-2013) Data

In order to optimise recall for the tsunami prediction model, the weight assigned to pre-2013 data was selected to be 0.05. This choice is based on a trade-off analysis: increasing the weight from 0.05 to 0.1 led to a sharp decline in recall, meaning the model missed more actual tsunami events. Although this change slightly improved AUC-ROC, it is insufficient to justify the sacrifice in recall. Between weights 0.4722 and 0.5778, recall and F1 not only plateaued but dipped into a trough, indicating a temporary decline in performance. This suggests that assigning higher weights to pre-2013 data in this range may have introduced noise or conflicting patterns, reducing the model's ability to detect tsunamis accurately. At the same time, AUC-ROC gains diminished past 0.3667, reinforcing that older data adds limited value and may even hinder performance when overweighted.

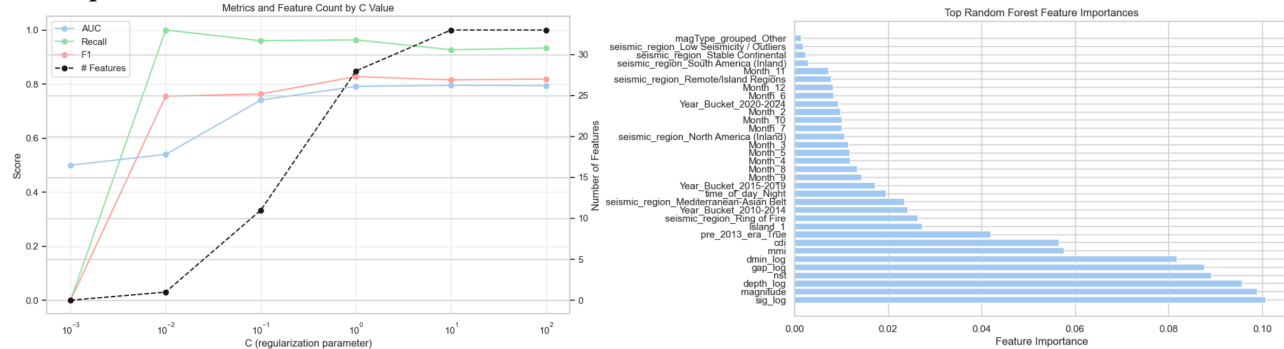


5.1.2 Optimal Resampling Method

When evaluated across different resampling weights, SMOTE consistently outperformed SMOTE-Tomek in terms of recall and AUC. The performance curve revealed that improvements in both recall and AUC began to slow beyond a SMOTE weight of 2, making it the optimal choice. At this elbow point, we achieved recall of 0.91 and a high AUC of 0.81, which limits introducing excessive false positives.



5.1.3 Optimal Features Set



Logistic Regression (L1) (left chart) achieved the highest cross-validation recall of 0.963. At the optimal regularisation strength ($C = 1.0$), it selected 28 features. Using forward selection (Appendix 6), we selected the 19-feature model by closely monitoring model performance at each step. Although AIC and BIC reached their minimum at just 5 features, we observed that such a small subset risked underfitting, as important predictors were likely excluded. Instead, we balanced performance with complexity by tracking recall, AUC, and F1 as additional criteria. At 19 features, the model achieved a high recall of 0.942 and strong AUC scores, alongside a significant dip in AIC/BIC score from the preceding iteration. Based on the Random Forest model's feature importance plot (right), we observed a significant drop in importance beyond the top 29 features, with the remaining features contributing negligibly. Therefore, we subset the top 29 features as the features selected by the Random Forest model. After triangulating the features selected by these methods, we found **13 features to be shared by all 3 methods**: 'pre_2013_era_True', 'magnitude', the encoded 'Month' variables for February, March, June, July, August, December, the encoded 'seismic_region' variables for the Mediterranean-Asian Belt and North America (Inland), 'cdi', 'Island_1' and 'depth_log'.

5.1.4 Overall Model Results

The table below highlights key models for comparison, with full results available in Appendix 7.

Model	Selected Features	CV Recall	CV AUC-ROC
Logistic Regression (L1)	28	0.963497	0.791422

Logistic Regression (L1, GridSearch)	13	0.993939	0.786110
Logistic Regression (L1, RandomizedSearch)		0.978788	0.787703
Random Forest (Default)		0.8421	0.8225
Random Forest (GridSearch)		0.987879	0.993650
Random Forest (RandomizedSearch)		0.984848	0.993242
KNN (GridSearch/ Randomized)		0.924242	0.921163
XGBoost (GridSearch)		0.990909	0.932685
XGBoost (RandomizedSearch)		1.000000	0.933735
SVM (GridSearch)		1.000000	0.783122

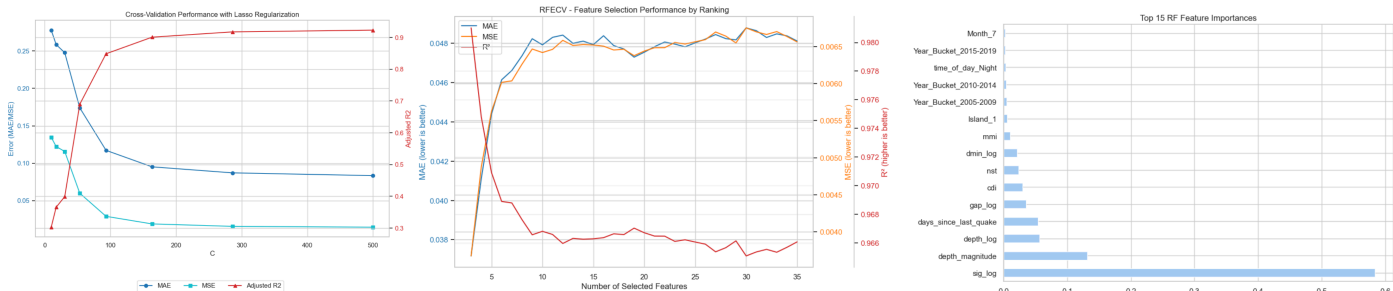
The improvement in the cross-validated recall for the Random Forest models (0.1) provides evidence that the triangulated features indeed managed to capture the most informative features and simplify the model, while enhancing its generalisability. **XGBoost (RandomizedSearch) emerged as the most effective model**, demonstrating the highest CV recall of 1 and AUC of 0.9337. Logistic regression, while less effective in terms of AUC, showed competitive recall and could still serve as a simpler model option, especially when interpretability is prioritised. Models such as KNN and SVM (Appendix 7) were less effective compared to the more sophisticated ensemble methods. KNN likely suffered due to the high dimensionality of the dataset, which can dilute the relevance of distance metrics used in nearest-neighbour calculations, leading to poor recall and AUC. In contrast, while SVM with kernel tuning achieved high recall, it showed poor overall accuracy and AUC. This suggests that it may have aggressively prioritised the positive class, resulting in excessive false positives. This reinforces the advantage of ensemble-based models, which are better equipped to manage complex decision boundaries. The final recommended model is XGBoost with RandomizedSearch derived parameters, which provided a Test recall value of 0.954545 and AUC of 0.754 (Appendix 9) .

5.1.5 Stress-Testing to Reflect Real-World Costs

Our cost-sensitivity analysis (Appendix 8) demonstrates that while the default 0.5 classification threshold is not optimised for specific false-negative/false-positive tradeoffs, it provides a robust baseline that performs reasonably across varying cost scenarios. The analysis shows that even at extreme cost ratios (e.g., FN:FP=50:1), the performance degradation from using 0.5 versus an optimised threshold is relatively modest, suggesting the model's discriminative power remains stable across decision boundaries. Importantly, the 0.5 threshold offers a balanced compromise when true error costs are unknown, avoiding both excessive false alarms (which could erode trust in the system) and dangerous under sensitivity (which could miss critical events). The smooth, predictable relationship between threshold adjustments and cost changes indicates that while optimal thresholds can provide incremental improvements for specific contexts, the default threshold serves as a practical choice that maintains reasonable performance across a wide range of potential cost asymmetries.

5.2 Regression: Earthquake Magnitude

5.2.1 Optimal Features Set



Lasso regression selected 10 features at the elbow point where 'C' = 93.51, where it achieves a good balance between model complexity and performance. At this 'C' value, the Lasso model's MAE was 0.117 and its MSE was 0.029, with an adj. R² of 0.849. Random Forest, on the other hand, highlighted 9 features with meaningful importance (> 0.01),

demonstrating more concentrated feature selection. While RFECV initially recommended only 3 features, we opted to retain the top 9, which is the point at which gains in all of the metrics start to plateau, to avoid underfitting. With RFECV, we achieved a lower MAE of 0.048 and MSE of 0.0065.

Since the number of selected features across Lasso, Random Forest importance scores, and RFECV was fairly consistent, we triangulated the results to identify stable predictors. Eight features were common across all three methods ('cdi', 'days_since_last_quake', 'depth_log', 'depth_magnitude', 'dmin_log', 'gap_log', 'mmi', 'nst', and 'sig_log'), which suggests strong agreement on their contribution as predictors. There were only a few unique features exclusive to Lasso and RFECV respectively, with 'Year_Bucket_2015-2019' appearing only in Lasso and 'Island_1' and 'Year_Bucket_2005-2009' unique to RFECV. Interestingly, both methods identified different year-based features, suggesting that temporal effects may be relevant but captured differently depending on the selection method. To evaluate whether these additional features improved generalisation, we tested three sets: (1) the 8 features common to all methods, (2) common features plus Lasso-only feature(s), and (3) common features plus RFECV-only feature(s). This allowed us to assess if the unique additions enhanced predictive performance or introduced unnecessary complexity.

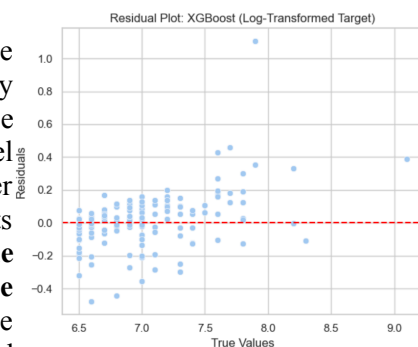
5.2.2 Overall Model Results

Model	Set 1: Triangulated Features			Set 2: Lasso Features			Set 3: RFECV		
Linear Regression	Train MAE	Train MSE	Train Adjusted R ²	Train MAE	Train MSE	Train Adjusted R ²	Train MAE	Train MSE	Train Adjusted R ²
Random Forest	0.078582	0.013574	0.928846	0.077983	0.013061	0.931426	0.079056	0.013379	0.929647
XGBoost	0.047586	0.006306	0.966945	0.055401	0.008697	0.954338	0.047976	0.006296	0.966890
SVR	0.025237	0.001179	0.993819	0.009885	0.000221	0.998840	0.025375	0.001230	0.993530
KNN	0.070549	0.016233	0.914907	0.069997	0.015912	0.916457	0.070448	0.016168	0.914981
XGBoost (Quantile)	0.000000	0.000000	1.000000	0.000000	0.000000	1.000000	0.000000	0.000000	1.000000
	0.043216	0.011322	0.940652	0.044632	0.010568	0.944513	0.024022	0.003599	0.981075

After evaluating the performance metrics across the three feature sets, **Feature Set 2 emerges as the best choice**. It demonstrates superior performance with the lowest MSE (0.000221) and the highest adj. R² (0.998840) in one of its entries, indicating an excellent balance between accuracy and model fit. While Feature Sets 1 and 3 also show strong results, Feature Set 2 consistently outperforms in key metrics. Therefore, Feature Set 2 is recommended as the optimal feature set for model training, offering the most reliable and robust performance. A textbook sign of severe overfitting is observed in several models. This is evident from their large discrepancies between training and test performance, which is further confirmed in Appendix 11. For instance, XGBoost (Quantile) demonstrates a strong overfit: while its training metrics are excellent (Train MAE = 0.0446, Train R² = 0.94), its Test MAE increases to 0.257 and Test R² drops to 0.357, indicating it does not generalize well to unseen data. A similar pattern is observed in the Random Forest model, where a relatively large train-test performance gap points to potential overfitting. XGBoost (Standard), while also fitting the training data very well, shows slightly better generalisation and the highest overall adj. R² on the test set, suggesting that it balances fit and generalisation better than the others. Notably, Linear Regression and SVR, despite being simpler models, show comparatively stable train-test performance and behave similarly in their residual plots (Appendix 10), where heteroskedasticity is present. This issue, where residual variance increases with the magnitude of the predicted value, is least pronounced in XGBoost. Overall, **XGBoost performs the best due to its strong performance and relatively well-behaved residuals**, benefiting from its non-parametric, tree-based structure that makes it less sensitive to heteroskedasticity.

5.2.3 Addressing Heteroskedasticity

After applying a log transformation to the target variable, we observed notable improvements in the model's residual behavior. While the transformation did not fully correct the right-skewness of the magnitude distribution (See 3.3), it reduced the influence of large-magnitude outliers. In particular, the residual plot of the log-transformed model showed a tighter spread of errors across different magnitude levels: a tighter lower (fewer overpredictions), and a lower upper bound (reduction in underprediction). This suggests that the log transformation helped the model generalise better. Overall, **the log-transformed model demonstrates improved stability and predictive performance across the target range**. Therefore, we will use the log-transformed target variable for the final model, which presents a test MAE of 0.1065, test MSE of 0.0277 and adjusted R-squared of 0.8539.



6. Discussion

6.1 Classification: Tsunami Occurrence

6.1.1 Model Interpretability

The SHAP analysis (Appendix 12) of our XGBoost tsunami prediction model reveals several scientifically interpretable patterns that validate its decision-making process. Most significantly, the model correctly identifies **earthquake magnitude and shallow depth (low ‘depth_log’ values) as the strongest positive predictors of tsunami risk**, perfectly aligning with established seismological principles about tsunamigenic earthquakes. The model demonstrates sophisticated data quality awareness by systematically downweighting pre-2013 records (‘pre_2013_era_True’=1), reflecting known limitations in seismic monitoring prior to two key milestones: the global deployment of DART buoys (2004) and subsequent seismic network upgrades (by 2013). This automatic data quality adjustment is particularly valuable as it mimics domain expertise while handling heterogeneous datasets. By automatically downweighting these older records, the model demonstrates context-aware interpretability, effectively compensating for historical data limitations that human experts would intuitively distrust. **Geographic features show appropriate directional impacts: island proximity (Island_1) and Mediterranean-Asian Belt locations increase predicted tsunami likelihood**, while inland North American earthquakes are appropriately discounted. The model correctly **ignores irrelevant human-reported intensity (cdi), focusing instead on physically meaningful signals**, though the minor influence of monthly features (e.g., Month_7) warrants further investigation to determine if these represent real seasonal patterns or data artifacts. This combination of physically-grounded feature importance (**magnitude > depth > location > data quality**) with contextual data quality adjustments makes the model both interpretable and operationally robust.

6.1.2 Future Work

Future work should focus on **cost-sensitive analysis to quantify the real-world implications** of false positives (unnecessary evacuations) versus false negatives (missed warnings), which would allow optimisation of decision thresholds for operational deployment scenarios where the cost asymmetry between error types may be substantial. Critically, these costs vary dramatically across populations: evacuation costs for dense urban coastal cities differ substantially from remote island communities, while the human and economic toll of missed warnings depends on local infrastructure and preparedness systems. This geographical and socioeconomic variability necessitates **region-specific threshold optimisation** when deploying the model operationally, particularly for vulnerable communities where traditional cost-benefit analyses may not adequately capture the true impact of prediction errors.

6.2 Regression: Earthquake Magnitude

6.2.1 Model Interpretability

While our final model, XGBoost, is typically considered a “black box” ensemble method, using SHAP values can provide crucial insights into feature importance and directional impacts, making the model’s decisions more transparent to seismologists, data scientists and decision makers. The analysis (Appendix 13) reveals that **depth-related features (‘depth_magnitude’ and ‘depth_log’) dominate the model’s predictions**, with SHAP values showing a clear physical interpretation: deeper earthquakes consistently yield lower magnitude predictions (-0.15 SHAP impact) while shallower events increase predicted magnitudes (+0.10). This model **corroborates seismological understanding**, demonstrating that the XGBoost model has learned physically meaningful patterns despite its complex architecture. The sig_log feature shows more nuanced bidirectional effects (± 0.05), suggesting the model may be capturing secondary signal characteristics that require domain expertise to fully interpret. Notably, human-reported intensity metrics (‘cdi’, ‘mmi’) and temporal features show near-zero SHAP values, confirming the model properly ignores these less relevant factors for magnitude prediction. This interpretability is particularly valuable for operational deployment, as it allows seismologists and emergency managers to understand and trust the model’s predictions. The analysis confirms that while XGBoost is complex, SHAP values successfully reveal its physically plausible reasoning, making it both powerful and transparent for high-stakes tsunami forecasting applications. Future refinements could incorporate additional geological parameters and cost-sensitive thresholds to optimize predictions for specific regional needs.

6.2.2 Future Work

We recommend incorporating additional geological parameters like fault dip and rupture area to enhance the physical basis of predictions, developing cost-sensitive or reasonably weighted regression approaches that account for the asymmetric consequences of over- versus under-prediction across different regions, and expanding the temporal data quality framework to address modern instrumentation gaps.

7 Conclusion

This study set out to apply predictive analytics to enhance early warning systems for seismic disasters, focusing on earthquake magnitude estimation and tsunami occurrence prediction. By integrating diverse data sources and employing advanced preprocessing techniques, we developed tailored machine learning models to support high-stakes decision-making in disaster preparedness. XGBoost, combined with resampling, proved to be the most effective classifier—achieving perfect recall while minimizing the cost of false positives. For magnitude regression, XGBoost delivered the strongest performance, making it a robust solution for anticipating seismic severity.

Despite strong results, several limitations remain. The classification model's high sensitivity led to increased false positives, which may incur economic costs. However, it is acceptable given the far greater cost of false negatives and the life-saving potential of precautionary responses. For regression, residual spread increased at higher magnitudes, highlighting continued challenges in modeling extreme events. Our exploration of threshold tuning based on hypothetical cost ratios further revealed that even small adjustments can substantially reduce expected cost. However, due to the lack of recent, reliable financial data, we opted not to rely on fixed dollar values. Future work could build on this by incorporating real-world cost data from government or insurance sources to develop more nuanced, context-specific cost functions and improve threshold optimization accordingly.

Nevertheless, the models developed in this study represent a valuable proof of concept. They offer a scalable, data-driven framework that, with further calibration and operational testing, could support disaster response agencies, insurers, and governments in proactive seismic risk management. This work not only advances the technical frontier of predictive disaster analytics but also lays the groundwork for more resilient and responsive early warning systems.

In future iterations, integrating real-time geospatial data and expanding temporal granularity could improve both prediction lead time and accuracy. Ultimately, our findings offer a data-driven framework that supports public agencies and insurers in making timely, informed decisions under seismic risk—laying the groundwork for scalable, real-world deployment of predictive tools in disaster response planning.

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Appendix

Appendix 1: Table of variables and their definitions

Kaggle data

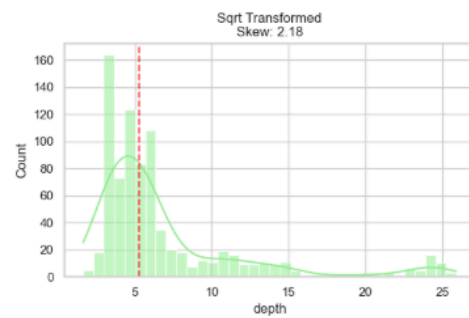
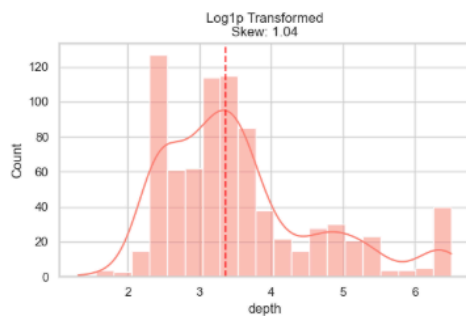
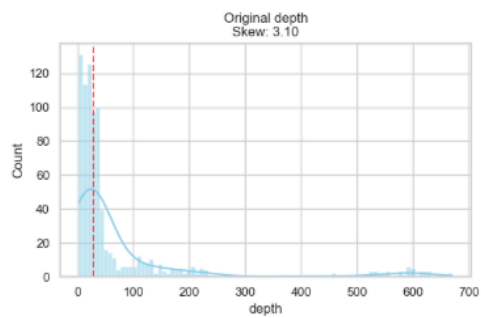
Variable	Data type	Definition	Transformations
magnitude	continuous	The magnitude of the earthquake	-
date_time	datetime	Date and Time	Split into “Year”, “Month”, “hour”, and “time_of_day”
cdi	continuous	The maximum reported intensity for the event range	-
mmi	continuous	The maximum estimated instrumental intensity for the event	-
tsunami	discrete	"1" for events in oceanic regions and "0" otherwise	-
nst	discrete	The total number of seismic stations used to determine earthquake location	-
continent	categorical	Continent of the earthquake hit country	-
country	categorical	Affected country	Replaced by “seismic_region”

NOAA Tsunami data

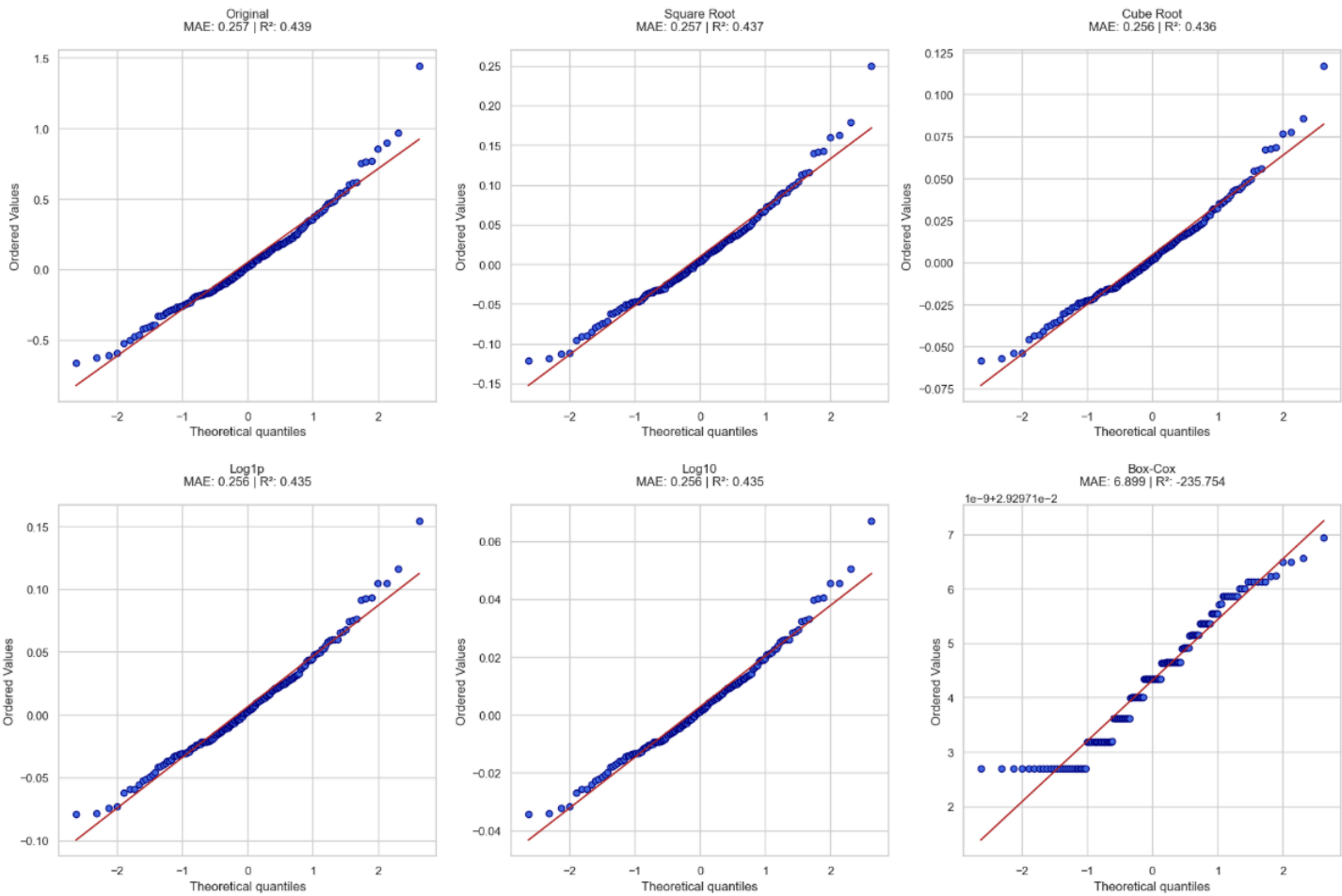
Variable	Data type	Definition	Transformations
Year	categorical	Year	Combined to create “datetime” column
Mo	categorical	Month	
Dy	categorical	Day	
Hr	categorical	Hour	
Mn	categorical	Minute	
Sec	categorical	Second	
Earthquake Magnitude	continuous	A continuous numerical variable ranging from 0.0 to 9.9, representing the energy released at the earthquake’s source. This value is derived from seismograph measurements using various magnitude scales, with the most reliable available scale selected as the primary value. For historical events prior to instrumental recordings, magnitudes are estimated based on reported intensities	-
Country	categorical	The Country where the tsunami source occurred	-
Location Name	categorical	The Country, State, Province or Island where the tsunami source occurred (	-

Appendix 2: Distribution Plots Pre- and Post-Transformation of Numerical Variables

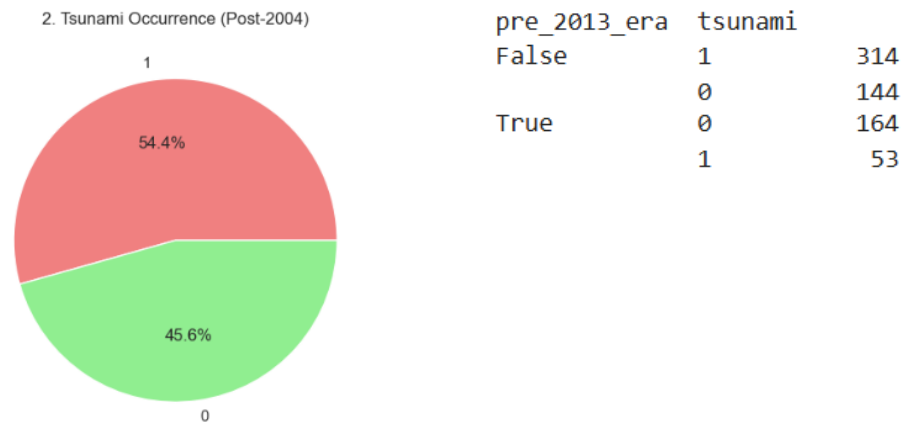




Appendix 3: Q-Q Plots of Transformations of ‘magnitude’ Variable

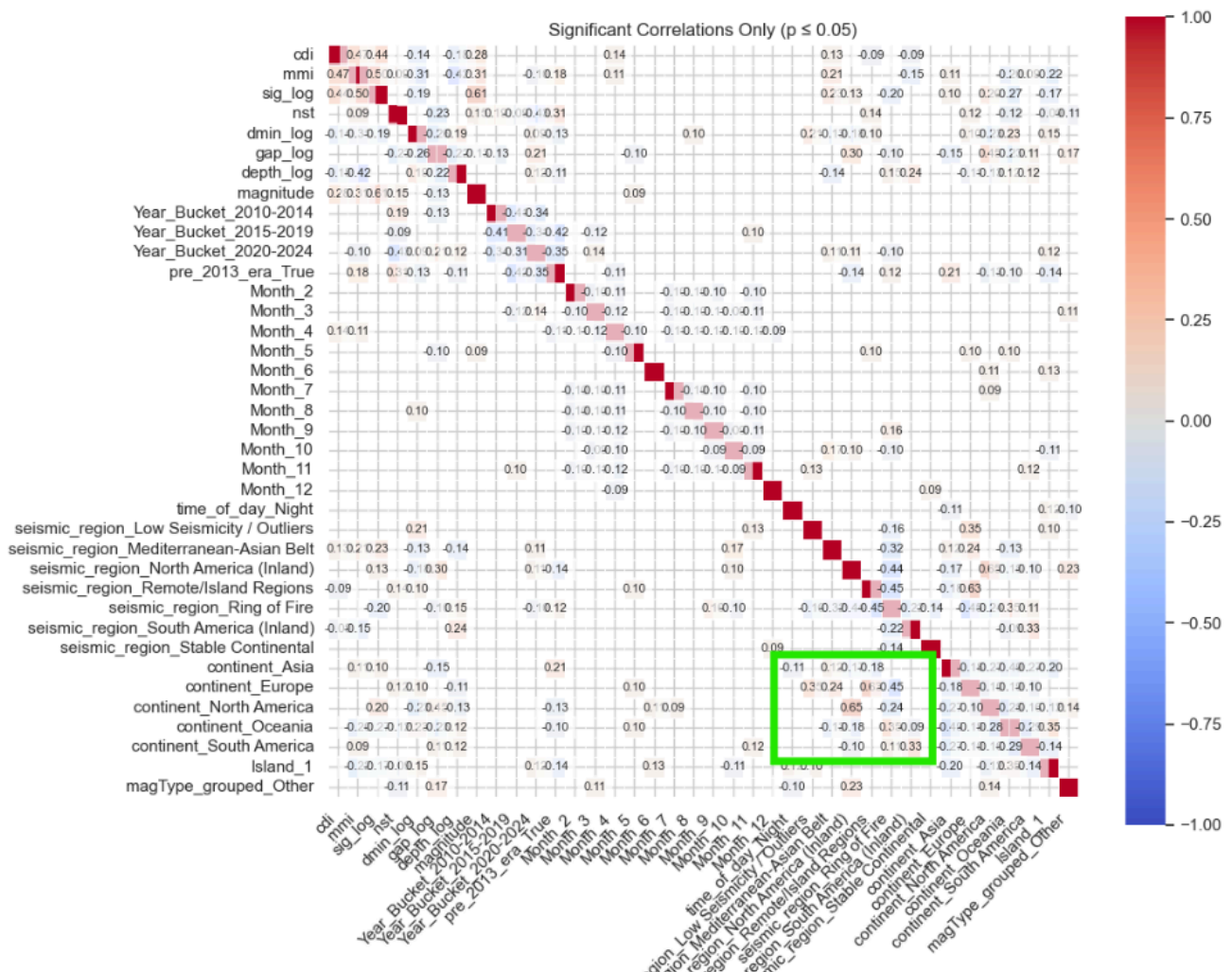


Appendix 4: Overall Class Distribution of ‘tsunami’ Label and Breakdown by Pre-2013 Data and Data Post-2012

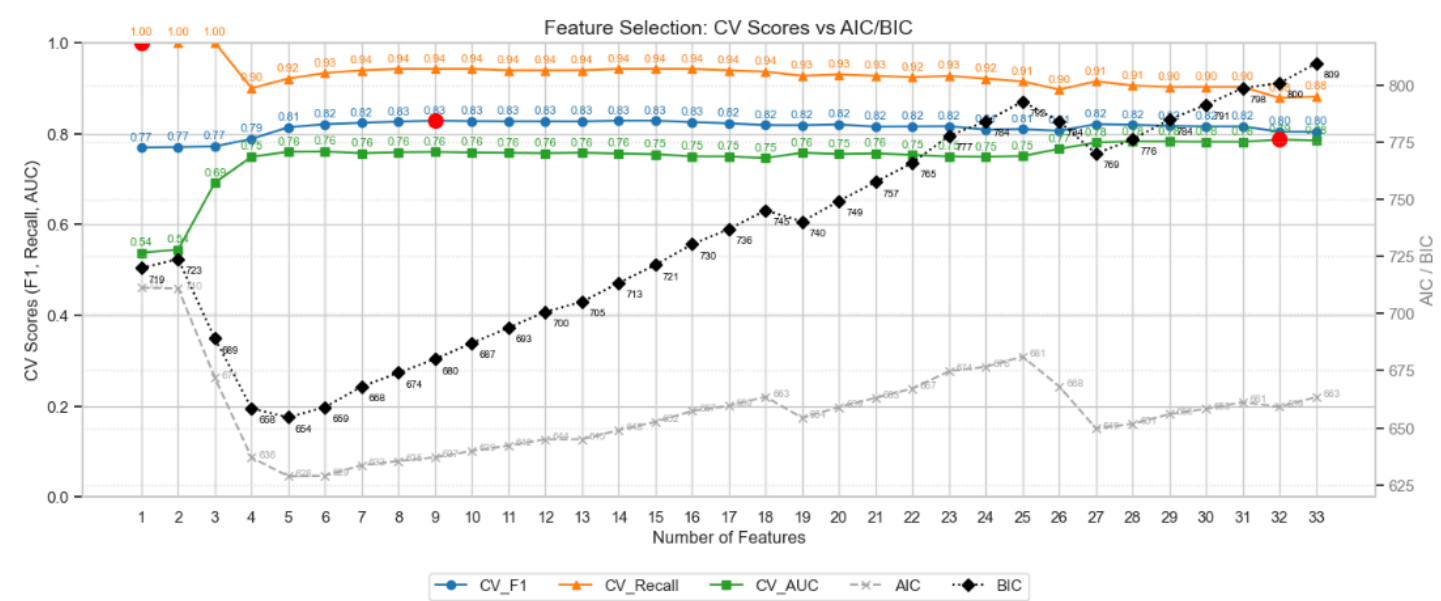


Appendix 5: Correlation Heatmap & Contingency Table of Encoded 'seismic_region' and 'continent' Variables

seismic_region	Himalayan Belt	Low Seismicity / Outliers	Mediterranean-Asian Belt	North America (Inland)	Remote/Island Regions	Ring of Fire	South America (Inland)	Stable Continental
continent								
Africa	0	0	0	0	0	0	0	3
Asia	40	0	10	0	0	158	0	0
Europe	0	4	10	0	29	0	0	0
North America	0	0	2	42	4	50	0	0
Oceania	0	0	0	0	14	225	0	1
South America	0	0	0	0	1	71	11	0



Appendix 6: Forward Selection Plot (Comparison of CV Scores, AIC and BIC)

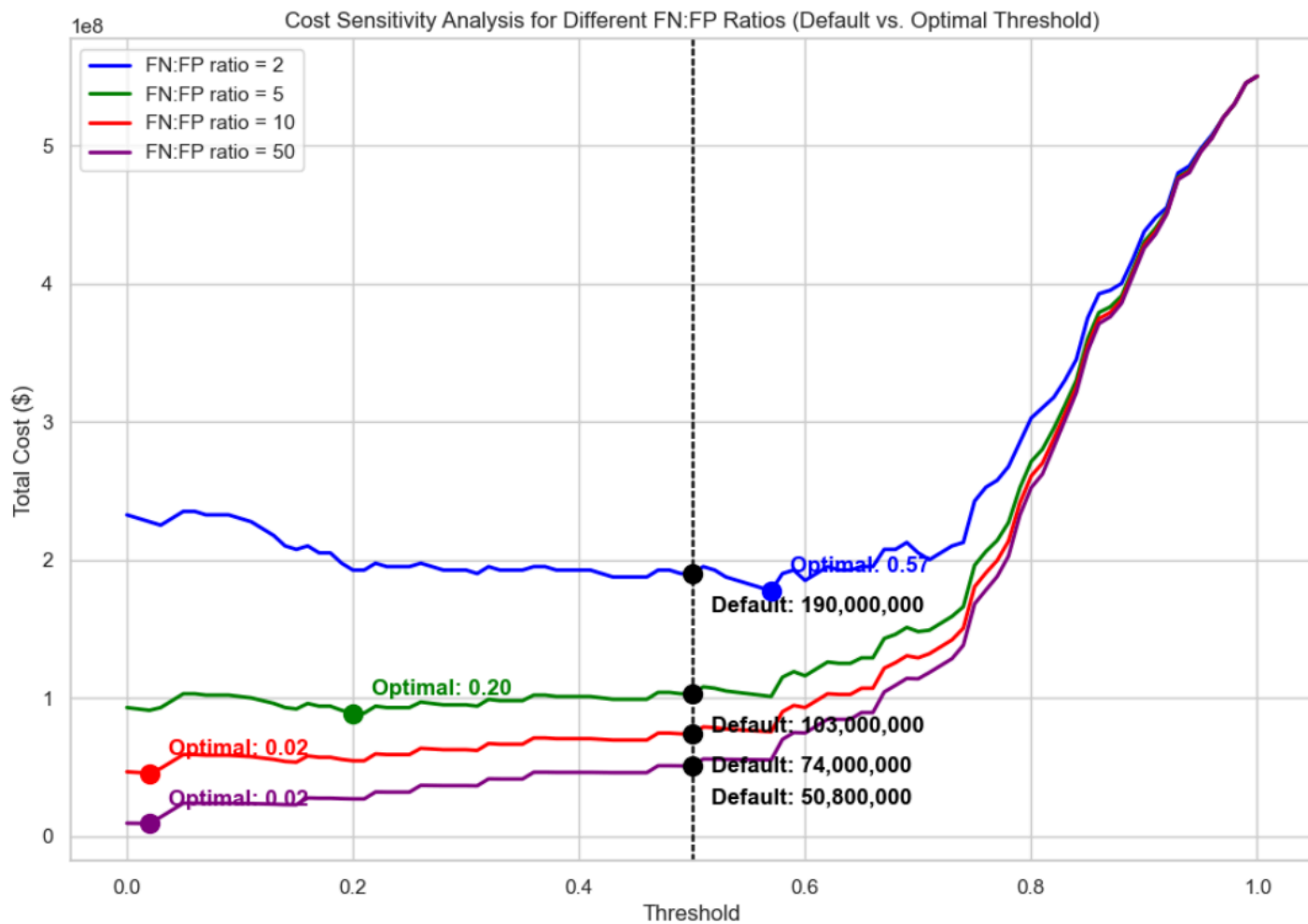


Appendix 7: Full Classification Model Results

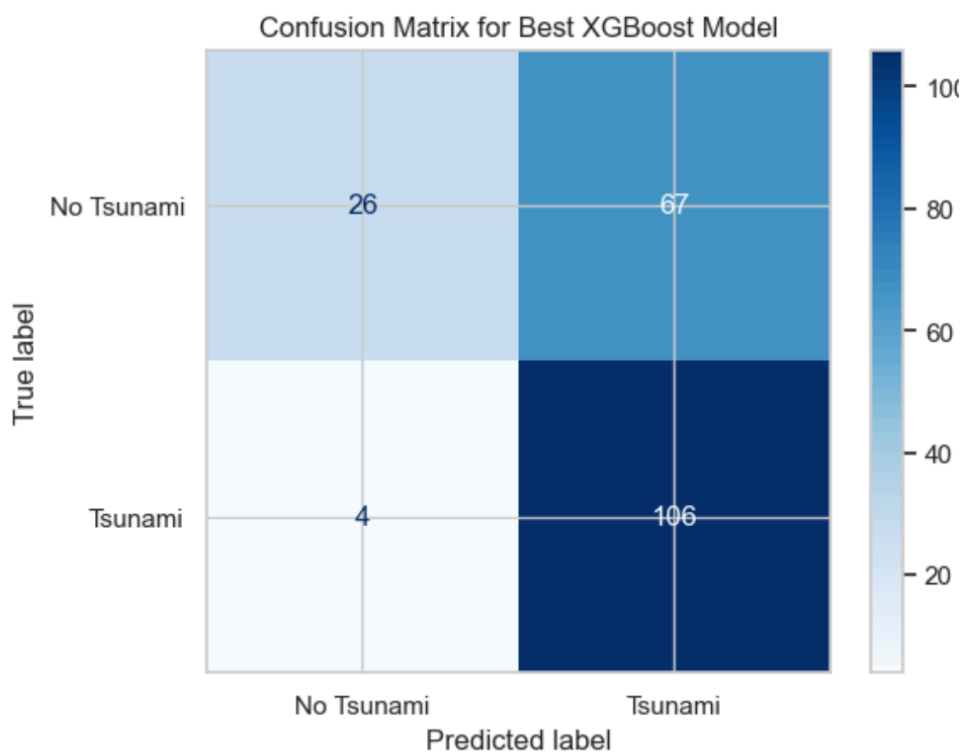
Model	Selected Features	CV Recall	CV AUC-ROC	Test Recall	Test AUC-ROC
Logistic Regression (Forward Selection)	19	0.930117	0.754869		
Logistic Regression (L1)	28	0.963497	0.791422		
Logistic Regression (L1, GridSearch)	13	0.993939	0.786110	0.963636	0.751369
Logistic Regression (L1, RandomizedSearch)		0.978788	0.787703	0.945455	0.741398
Random Forest (Default)		0.8421	0.8225		
Random Forest (GridSearch)		0.987879	0.993650	0.827273	0.745601
Random Forest (RandomizedSearch)		0.984848	0.993242	0.827273	0.742571
KNN (GridSearch)		0.924242	0.921163	0.736364	0.683920
KNN (RandomizedSearch)		0.924242	0.921163	0.736364	0.683920
XGBoost (GridSearch)		0.990909	0.932685	0.918182	0.756891
XGBoost		1.00000	0.933735	0.954545	0.754106

(RandomizedSearch)		0			
SVM (GridSearch)		1.00000 0	0.783122	0.981818	0.729472
SVM (RandomizedSearch)		0.99697 0	0.783080	0.945455	0.741398

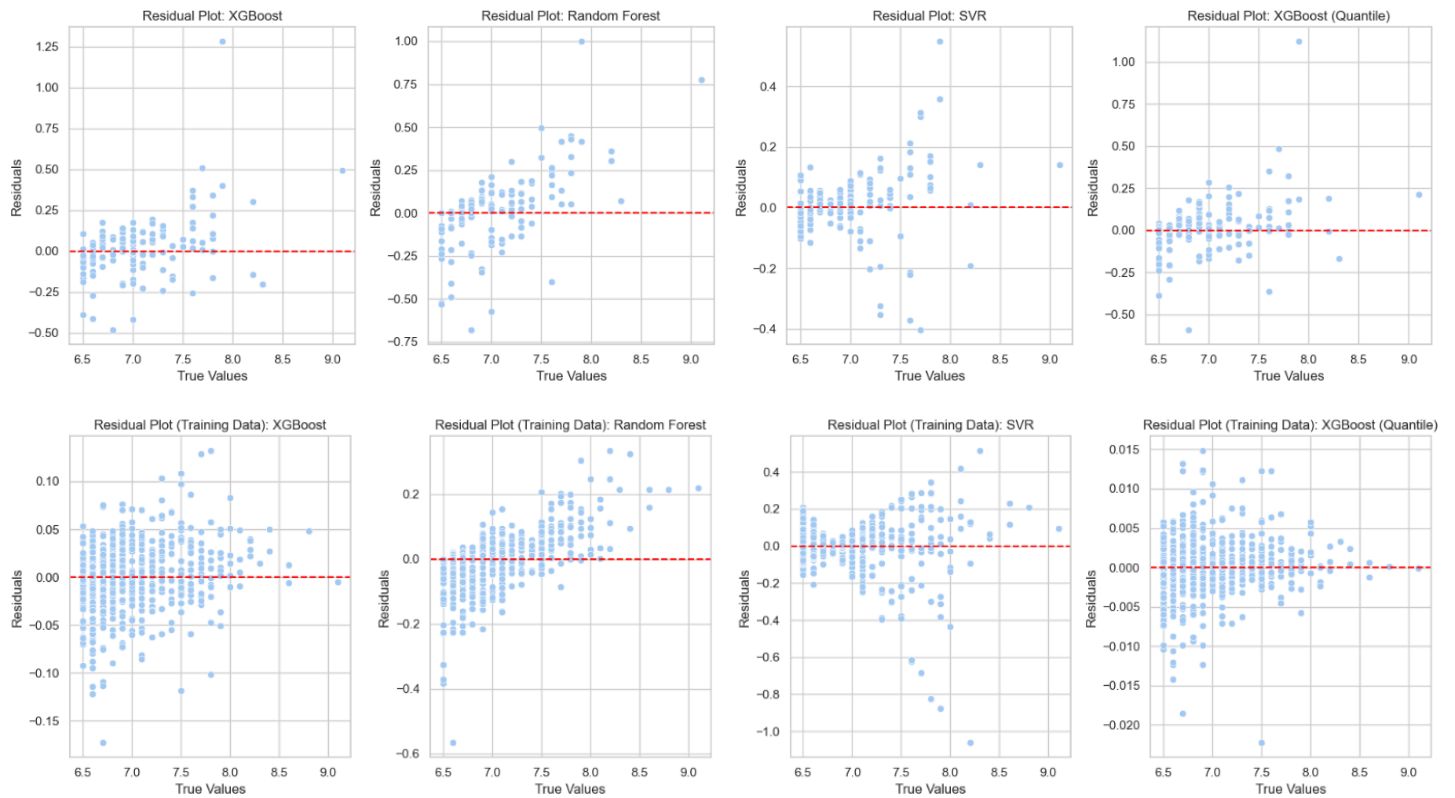
Appendix 8: Cost Sensitivity Analysis for Different FN:FP Cost Ratios



Appendix 9: Confusion Matrix of Final Model



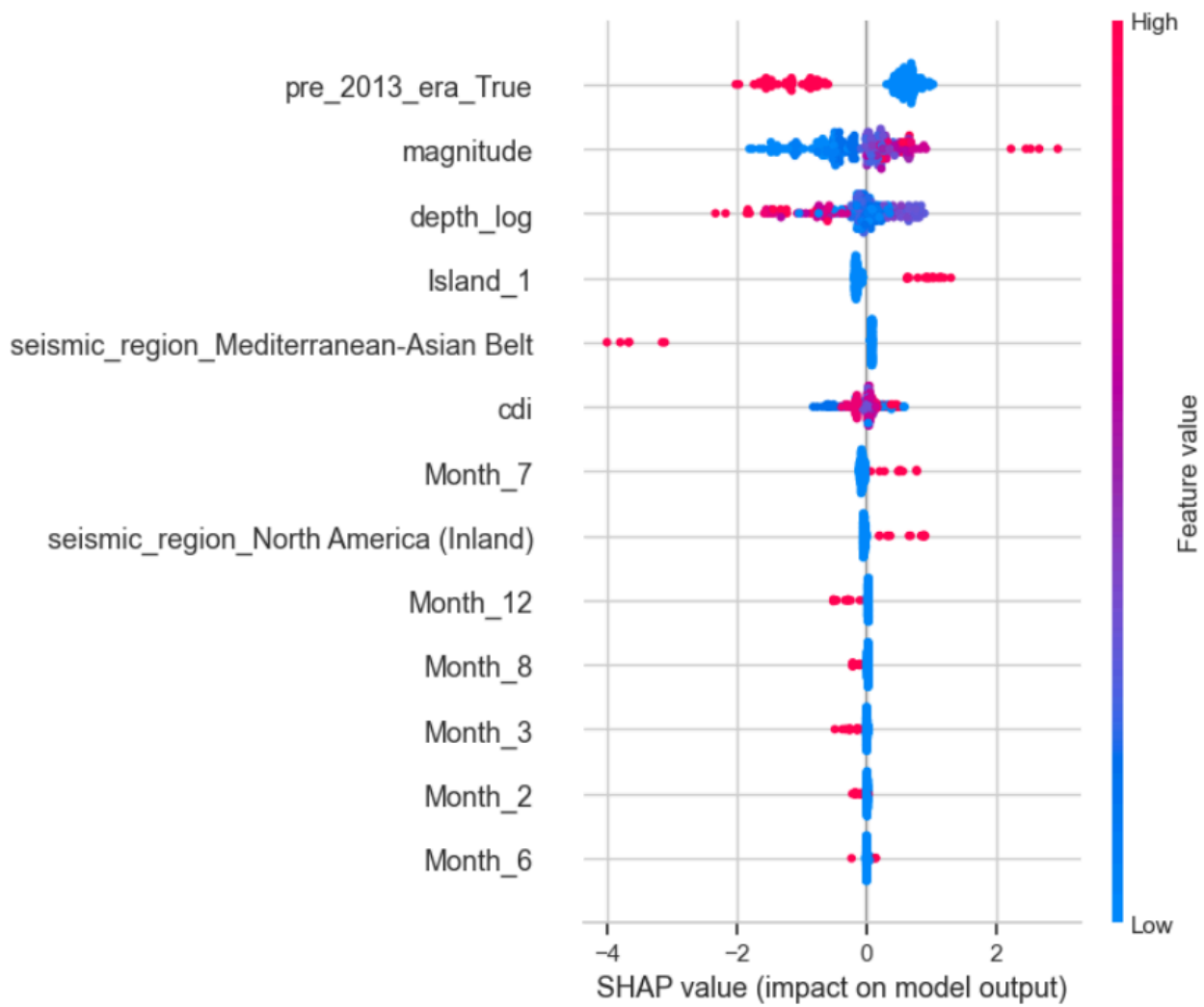
Appendix 10: Train and test residual plots



Appendix 11: Comparison of Train and Test Results with Different Feature Sets

Model	Set 1: Triangulated Features			Set 2: Lasso Features			Set 3: RFECV		
Linear Regression	Train MAE	Train MSE	Train Adjusted R ²	Train MAE	Train MSE	Train Adjusted R ²	Train MAE	Train MSE	Train Adjusted R ²
Random Forest	0.078582	0.013574	0.928846	0.077983	0.013061	0.931426	0.079056	0.013379	0.929647
XGBoost	0.047586	0.006306	0.966945	0.055401	0.008697	0.954338	0.047976	0.006296	0.966890
SVR	0.025237	0.001179	0.993819	0.009885	0.000221	0.998840	0.025375	0.001230	0.993530
KNN	0.070549	0.016233	0.914907	0.069997	0.015912	0.916457	0.070448	0.016168	0.914981
XGBoost (Quantile)	0.000000	0.000000	1.000000	0.000000	0.000000	1.000000	0.000000	0.000000	1.000000
	0.043216	0.011322	0.940652	0.044632	0.010568	0.944513	0.024022	0.003599	0.981075
Linear Regression	Test MAE	Test MSE	Test Adjusted R ²	Test MAE	Test MSE	Test Adjusted R ²	Test MAE	Test MSE	Test Adjusted R ²
Random Forest	0.079675	0.012169	0.936137	0.082296	0.012596	0.933458	0.081024	0.012390	0.934106
XGBoost	0.134795	0.044115	0.768488	0.138639	0.046812	0.752697	0.134636	0.045519	0.757917
SVR	0.108021	0.031300	0.835740	0.102381	0.030671	0.837968	0.102175	0.024694	0.868669
KNN	0.070793	0.012468	0.934568	0.071212	0.012515	0.933886	0.071306	0.012551	0.933252
XGBoost (Quantile)	0.242216	0.105158	0.448143	0.257080	0.121746	0.356834	0.251602	0.123588	0.342721
	0.126088	0.039868	0.790779	0.123602	0.038794	0.795058	0.117738	0.038165	0.797026

Appendix 12: SHAP Analysis (Classification)



Appendix 13: SHAP Analysis (Regression)

